RAJALAKSHMI ENGINEERING COLLEGE	**RAJALAKSHMI ENGINEERING COLLEGE** **B.E. COMPUTER SCIENCE AND ENGINEERING** **(CYBER SECURITY)** **CR23721 - PROJECT ABSTRACT (2026-27)**

TITLE	**AI-Based Fingerprint and Iris Privacy Protection System**
TEAM MEMBERS **REG. NUM &NAME**	231901043 – RITHUPRIYA S 231901048 – SHAMRUDHA VARSHINI P
PROJECT GUIDE	Dr Bhuvana S Assistant Professor(SG)
TEAM NUMBER	11

ABSTRACT:

The rapid growth of social media has increased the sharing of high-resolution images, leading to the unintentional exposure of sensitive biometric information such as fingerprints and iris patterns. Such exposure can result in identity theft, biometric spoofing, and privacy violations. Existing biometric systems mainly focus on authentication and recognition, whereas limited attention has been given to protecting biometric information before images are shared online.This project proposes an AI-Based Fingerprint and Iris Privacy Protection System that analyzes uploaded images to identify biometric exposure before they are posted on social media. This system uses OpenCV for image preprocessing, MediaPipe Hands and MediaPipe Face Mesh for fingerprint and iris region detection, and a Vision Transformer (ViT) deep learning model to assess the biometric privacy risk. Based on the analysis, the system classifies the image into Low, Medium, or High privacy risk levels and provides appropriate recommendations such as fingerprint blurring and iris masking .A web-based system is developed to simulate a social media platform where images are scanned for biometric privacy risks before upload. By providing real-time privacy alerts and protection recommendations, the proposed system helps users safeguard their biometric information and promotes secure image sharing. This project combines Artificial Intelligence, Computer Vision, and Cybersecurity to provide a proactive solution for biometric privacy protection in digital platforms.

PROJECT GUIDE **PROJECT COORDINATOR**